先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 06 – Feature Engineering: Transformation, Construction, and Selection



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Measurements must be available when we predict

- To predict illness at admission, use measurements already available at admission.
- A discharge diagnosis reveals information from the future.
- Check what every column means and when it was recorded.

Syllabus allocation for Lecture 06

Syllabus section	Time	What students must do
Feature transformation	1.0 hour	fit and apply valid representations
Feature construction	0.5 hour	turn domain hypotheses into variables
Feature selection	1.5 hours	compare filter, wrapper, and embedded methods

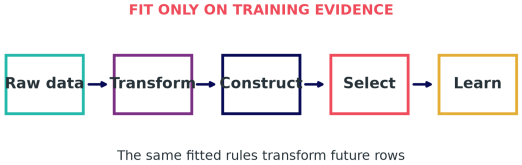


Course boundary

Principal component analysis belongs to the later dimensionality reduction lecture.

Feature engineering sits between raw data and learning

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Raw data. Measurements, categories, text, time stamps, and missingness.

Feature pipeline. Transform, construct, and select using training data only.

Learner. Fit a model and evaluate it on untouched evidence.

Feature type determines valid operations

Raw type	Examples	Valid first questions
Continuous	temperature, price	scale, skew, outliers
Ordinal	pain level, grade	order and spacing
Nominal	ward, color	encoding and unknown values
Time	date, hour	cycle and prediction cutoff

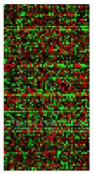


face images

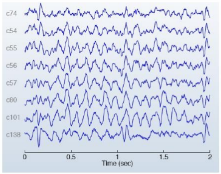
Zambian President Levy Mwanawasa has won a second term in office in an election his challenger Michael Sata accused him of rigging, official results showed on Monday.

According to media reports, a pair of hackers said on Saturday that the Firefox Web browser, commonly perceived as the safer and more customizable alternative to market leader Internet Explorer, is critically flawed. A presentation on the flaw was shown during the ToorCon hacker conference in San Diego.

documents



gene expression data



MEG readings

Fit and transform are different actions

Fit

Learn parameters such as means, medians, category sets, or bin edges from training data.

Transform

Apply the learned rule without recalculating it on validation or test data.

Leakage

Calling fit on the test set lets future evidence influence the representation.

Split first, then fit preprocessing

- Split the raw examples into training, validation, and test sets.
- Use `fit(X_train)` to learn means, scales, or categories.
- Use `transform` with the same fitted rule for all sets.
- Inside cross-validation, repeat fitting within each training fold.

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Missingness has several mechanisms

Mechanism	Plain meaning	Risk
Missing completely at random	unrelated to observed or hidden values	mostly lost information
Missing at random	explained by observed variables	model the observed context
Missing not at random	depends on the missing value or hidden cause	simple imputation can bias

Row and column deletion solve different problems

Drop rows

Reasonable when missingness is rare and approximately random.

Drop columns

Reasonable when a field is mostly unavailable or operationally unreliable.

Selection bias

Deleting rows can remove a specific patient group rather than random noise.

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Mean, median, and mode encode assumptions

Imputer	Suitable situation	Failure
Mean	roughly symmetric numeric data	moves with outliers
Median	skewed numeric data	ignores subgroup structure
Mode	categorical or discrete data	can exaggerate the majority



Imputation learns training statistics

$$\tilde{x}_{ij} = \begin{cases} x_{ij}, & x_{ij} \text{ observed} \\ m_j, & x_{ij} \text{ missing} \end{cases}$$

Training estimate

m_j is learned from the training fold only.

Test time

Reuse the same value even when the new batch has a different median.

A missing indicator can preserve process information

Construct

$m_j = 1$ when feature j is missing and 0 otherwise.

Clinical process

A test was not ordered because the initial examination looked normal.

Caution

The reason for missingness may change when hospital policy changes.

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A histogram comes before a transformation

Inspect

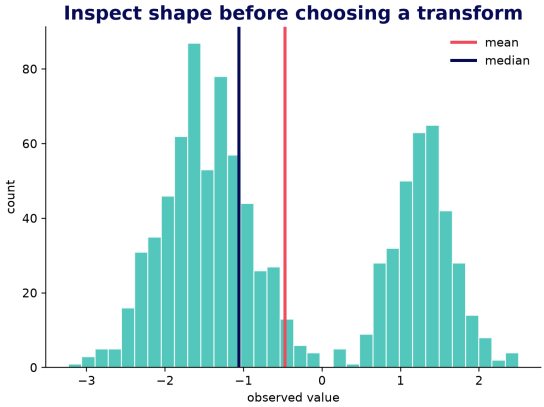
Look for skew, multiple modes, truncation, and impossible values.

Question

Does one peak represent one population or two mixed subgroups?

Do not beautify blindly

A smoother distribution is not automatically more predictive.



Outliers may be errors or rare truths

Error

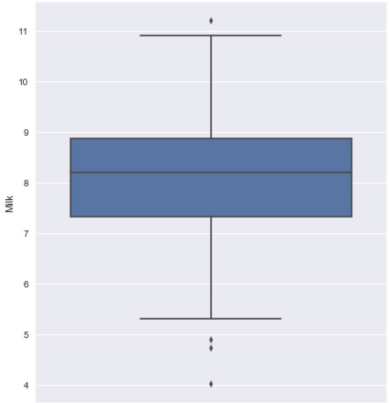
Body temperature recorded as 370 instead of 37.0.

Rare truth

An unusually high transaction that is a genuine fraud signal.

Decision

Validate the record and its units before clipping or removing it.



Standardization centers and rescales each feature

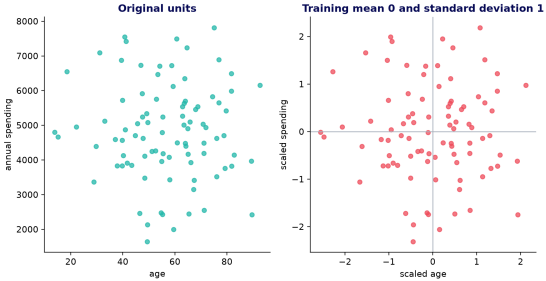
$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

Training parameters

Estimate μ_j and σ_j from the training fold.

Interpretation

One unit of z means one training standard deviation.



Min max scaling fixes a training range

$$x_j' = \frac{x_j - \min_{\text{train}} x_j}{\max_{\text{train}} x_j - \min_{\text{train}} x_j}$$

Training values

Map the observed training minimum to 0 and maximum to 1.

Future value

A new value outside the training range can map below 0 or above 1.

Robust scaling uses quantiles

$$x_j' = \frac{x_j - \text{median}(x_j)}{Q_{0.75}(x_j) - Q_{0.25}(x_j)}$$

Benefit

Median and interquartile range react less to extreme observations.

Tradeoff

Quantiles still come from training data and can shift across populations.

Row normalization answers a different question

$$x_i' = \frac{x_i}{\|x_i\|_2}$$

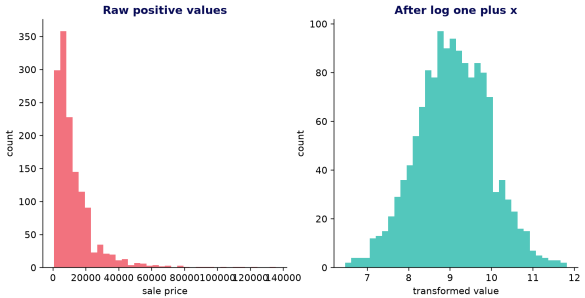
Across columns

Standardization rescales each feature using many samples.

Across one row

Normalization rescales one sample vector to unit length.

Log transforms compress right tails



$$x' = \log(1 + x)$$

Effect. Large values are compressed more than small values.

Use case. Price, counts, and response times often span several orders of magnitude.

Power transforms include log as one case

Family

Box Cox for positive values and Yeo Johnson for values that may include zero or negatives.

Learned parameter

The power parameter must be fitted inside the training fold.

Interpretation

Automatic Gaussian looking output does not guarantee better prediction.

Binning turns a continuum into intervals

Age

18 to 25, 26 to 40, 41 to 60, and above 60.

Benefit

A linear model can represent step changes and saturation.

Cost

Values near a boundary can receive abruptly different codes.

Quantile bins must be learned on training data

Equal frequency idea

Choose bin edges so training bins contain similar counts.

Leakage

Computing quantiles on all rows lets validation and test distributions set the boundaries.

Deployment

Future values use the stored training edges even when bin counts become unequal.

Ordinal and nominal categories need different encodings

Type	Example	Possible encoding
Ordinal	low, medium, high	ordered integer with verified order
Nominal	red, green, blue	one hot or another order free encoding

Distance

An integer code for nominal values invents an order and spacing.

One-hot encoding represents categories without ranking them

- Red $\mapsto (1, 0, 0)$; green $\mapsto (0, 1, 0)$; blue $\mapsto (0, 0, 1)$.
- Codes 1, 2, and 3 can imply an order or distance that colors do not have.
- Learn the category vocabulary from training data.
- Define how to handle an unseen category at prediction time.

Dropping one indicator depends on the model

Linear model with intercept

One category column may be redundant because all indicators sum to one.

Regularized or tree model

Keeping all columns may be acceptable and easier to interpret.

Reference meaning

The dropped category becomes the baseline for coefficient comparisons.

Unknown categories need a deployment policy

Training

Ward codes A, B, and C are observed.

Future

A new ward D appears after reorganization.

Policy

Ignore into an all zero code, map to other, or retrain under a controlled update.

High cardinality makes one hot sparse

Examples

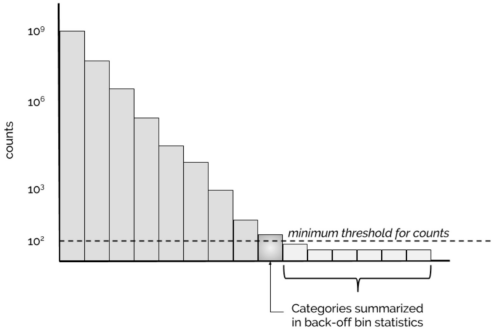
User identifier, product identifier, postal code, and device identifier.

Statistical problem

Many categories have too few examples to estimate reliable effects.

Computational problem

The design matrix becomes very wide and sparse.



Frequency encoding uses only feature counts

$$e(c) = \frac{\#\{i : x_i = c\}}{N}$$

Advantage

No target label is used and one column can replace many indicators.

Collision

Different categories with the same frequency receive the same value.

Target encoding can leak labels

$$e(c) = \mathbb{E}[y \mid x = c]$$

Temptation

Replace each category with its observed training target rate.

Self inclusion

A rare row can partially encode its own label.

Target encoding can leak the answer

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信息泄漏

- 分箱计数策略中使用了标签y的信息，容易发生信息泄漏。
- 一种防漏的方式是使用历史数据进行统计，这些统计信息用于当前的特征编码。
- 实践中，增加一个拉普拉斯分布的小随机噪声，可以掩盖单个数据点潜在的泄漏。

class sklearn.preprocessing.**TargetEncoder**(***categories***='auto', ***target_type***='auto', ***smooth***='auto', ***cv***=5, ***shuffle***=True, ***random_state***=None)

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Date time fields contain several clocks

- Calendar: year, month, weekday, and holiday.
- Clock: hour and minute.
- Elapsed time: age of an event or time since the previous event.
- Cutoff: information available before prediction.

Raw timestamp

A large epoch number rarely expresses the right relationships by itself.

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Encode time on a circle

- For hour h, use $\sin(2\pi h/24)$ and $\cos(2\pi h/24)$.
- Hours 23 and 0 are close on this circle.
- A single sine value is insufficient: two hours can share it.

Text and images need feature extraction

Text

Counts, term weighting, or learned embeddings convert tokens into numeric representations.

Images

Pixels, engineered descriptors, or learned representations become feature vectors.

Course boundary

Detailed representation learning appears in the deep learning lectures.



A mixed column transformer in Python

```
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler

numeric = make_pipeline(SimpleImputer(strategy="median"),
                        StandardScaler())
categorical = make_pipeline(SimpleImputer(strategy="most_frequent"),
                            OneHotEncoder(handle_unknown="ignore"))
prep = ColumnTransformer([
    ("num", numeric, numeric_columns),
    ("cat", categorical, categorical_columns),
])
```

Contract

Calling fit on the full pipeline learns every preprocessing parameter from training data.

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Build a feature that answers a useful question

- Price per unit: total price divided by quantity.
- Time since last purchase: prediction time minus last purchase time.
- Missing-value indicator: was a measurement recorded?
- Check units, zero denominators, and availability at prediction time.

A taxi estimate starts with a map question

Raw inputs

Pickup latitude, pickup longitude, dropoff latitude, dropoff longitude, and departure time.

Useful relation

Travel distance and traffic period are closer to the mechanism of trip duration.

Unavailable value

Actual arrival time cannot be used when the estimate is made.

Distance combines four coordinates

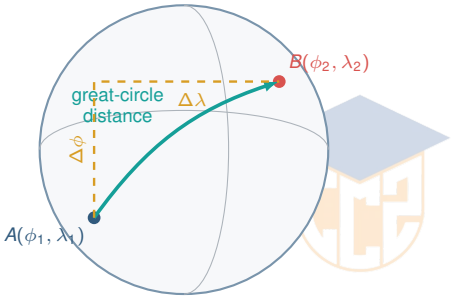
$$d = 2R \arcsin\left(\sqrt{\sin^2\frac{\Delta\phi}{2} + \cos\phi_1 \cos\phi_2 \sin^2\frac{\Delta\lambda}{2}}\right)$$

Interpretation

Four geographic coordinates become one approximate great circle distance.

Model limits

Road distance can differ because of rivers, bridges, and street layout.



coordinates define the spherical arc; road distance may be longer

Ratios reveal scale free behavior

$$\text{occupancy} = \frac{\text{used capacity}}{\text{total capacity}}$$

Hospital beds

Forty occupied beds mean different pressure in a 50 bed and a 500 bed hospital.

Denominator

Near zero denominators create unstable or extreme ratios.

Differences encode change

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$$\Delta x = x_{\text{now}} - x_{\text{baseline}}$$

Patient monitoring

A temperature rise from a personal baseline can matter even when the absolute value is moderate.

Baseline

A baseline measured after the event would leak future information.

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An interaction combines two features

特征组合

- 某些特征经过关联之后，与标签之间的相关性就会提高
  - 例：将国家和假日组合
    - “ USA”与“Thanksgiving”
    - “China”与“Chinese New Year”
- 这样的关联特征，对用户的点击有着正向的影响
  - 来自“China”的用户很可能会在“Chinese New Year”有大量的浏览、购买行为，在“Thanksgiving”却不会有特别的消费行为。
- 更多示例：
  - “化妆品”类商品与“女”性
  - “球类运动配件”的商品与“男”性

Polynomial features bend a linear boundary

$$(x_1, x_2) \mapsto (x_1, x_2, x_1^2, x_1 x_2, x_2^2)$$

Benefit

A linear learner in the expanded space can represent curved relations.

Growth

Degree and dimension make the number of terms rise quickly.

Feature count grows before information grows

$$p_{\leq d} = \binom{p+d}{d} - 1$$

Ten inputs

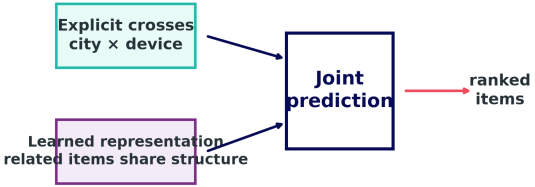
Degree three expansion produces 285 nonconstant terms.

Sample size

More columns do not create more independent observations.

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Wide and deep models use two kinds of memory



Wide part. Memorizes explicit crosses such as city and device combinations.

Deep part. Learns distributed representations that can generalize to related combinations.

Recommendation. Known purchase patterns and unseen but similar items need different strengths.

Elapsed time is often more useful than a date

$$\text{tenure} = t_{\text{prediction}} - t_{\text{start}}$$

Subscription

Account age carries a direct behavioral meaning that a signup date does not.

Negative values

A negative duration usually signals a timestamp error or leakage.

Rolling summaries need a closed historical window

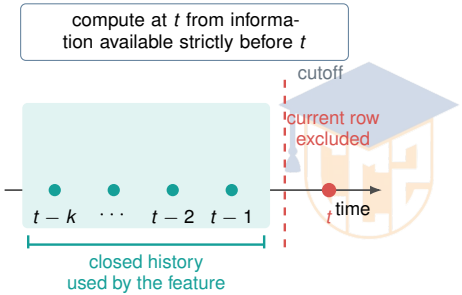
$$m_i(t) = \frac{1}{k} \sum_{s=t-k}^{t-1} x_i(s)$$

Fraud

Count transactions in the seven days before the current payment.

Current row

Including the event being predicted can reveal part of the answer.



Group aggregates can identify context or identity

Context

A store's historical average demand can describe its operating environment.

Identity shortcut

A nearly unique group statistic may let the model memorize an entity.

Cold start

A new store has no personal history.

Future leakage survives elegant code

```
orders = orders.sort_values("time")
orders["past_mean"] = (
    orders.groupby("customer")["amount"]
          .transform(lambda s: s.shift(1).rolling(10).mean())
)
```

Critical operation

shift(1) excludes the current order before the rolling mean is computed.

Feature hashing trades memory for collisions

$$j = h(c) \bmod m$$

Benefit

Map a huge or open category set into a fixed number of columns.

Collision

Different categories can share a column and their effects become entangled.

Missingness can interact with context

$$z = \mathbf{1}(x \text{ missing}) \times \mathbf{1}(\text{night shift})$$

Operations

A missing lab result at night may reflect staffing rather than patient physiology.

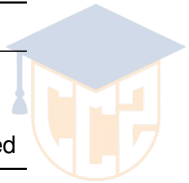
Policy drift

The interaction can change when staffing or ordering rules change.

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Units belong in the feature definition

Feature	Bad definition	Operational definition
speed	distance divided by time	kilometres divided by hours
dosage	dose divided by weight	milligrams divided by kilograms
price change	new minus old	currency and time window stated



A custom transformer makes a hypothesis reproducible

```
from sklearn.base import BaseEstimator, TransformerMixin

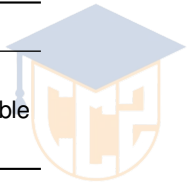
class RatioFeature(BaseEstimator, TransformerMixin):
    def fit(self, X, y=None):
        return self
    def transform(self, X):
        out = X.copy()
        out["occupancy"] = out["used"] / out["capacity"].clip(lower=1)
        return out
```

Benefit

The same formula becomes part of fitting, cross validation, and deployment.

Constructed features need ablation evidence

Model input	Validation score	Decision
raw fields	baseline	keep as reference
raw plus proposed feature	compare across folds	keep only if gain is stable
proposed feature alone	diagnostic	check for shortcuts



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What feature construction should achieve

- Convert categorical values without inventing an order.
- Represent time cycles and useful feature interactions.
- Fit every learned transformation within the training data.

Feature selection keeps existing columns

- A filter scores each feature before fitting the final predictor.
- Examples: variance, correlation, chi-square, or mutual information.
- A wrapper compares subsets using a fitted predictor.
- An embedded method selects features while fitting the model.

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Why remove features

- Reduce variance when samples are limited.
- Lower measurement and deployment cost.
- Shorten training and inference time.
- Remove redundant or unstable signals.
- Improve human review when selected variables retain meaning.

Myth

A smaller set is not automatically more accurate.

High dimension makes neighborhoods sparse

$$\frac{V_{\text{inner cube}}}{V_{\text{unit cube}}} = r^p$$

Scale

At r=0.8, the fraction is 0.8 in one dimension but about 0.012 in twenty dimensions.

Consequence

Local similarity needs far more data as dimension grows.

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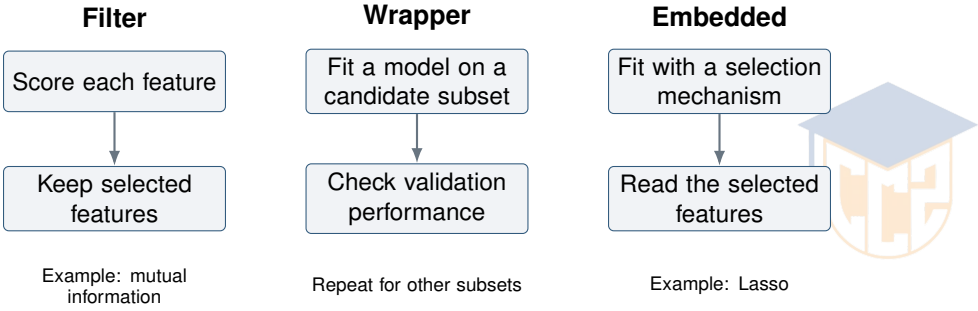
Selection and extraction preserve different things

Operation	Output	Meaning
feature selection	subset of original columns	original semantics remain
feature extraction	new combinations or coordinates	semantics can become mixed
feature construction	new domain variables	semantics are designed explicitly

Later lecture

Principal component analysis is treated under dimensionality reduction.

Filter, wrapper, and embedded feature selection



Fit selection inside the training folds. Using validation or test labels to preselect features leaks information.

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Fit selection separately inside each validation fold

- For each fold, fit the selector using that fold's training labels.
- Transform its training and validation rows with that fitted selector.
- Fit the predictor on transformed training rows; score validation rows.
- After choosing settings, refit the whole pipeline on all training data.

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A validation score can be optimistic before modeling

Thought experiment

Generate ten thousand random columns and keep the one most correlated with a random label.

Result

The chosen noise column looks related by chance on the same data.

Repair

Repeat the entire search inside training folds and test only once at the end.

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Variance threshold removes constants

$$\widehat{\mathrm{Var}}(x_j) < \tau \quad \Rightarrow \quad \text{remove } x_j$$

Use

Discard constant or nearly constant columns without using the target.

Scaling

The numerical threshold depends on feature units and encoding.

Univariate filters score one feature at a time

$$s_j = \text{score}(x_j, y)$$

Examples

Correlation, analysis of variance, chi square, and mutual information.

Blind spot

A weak individual feature can become useful through an interaction.

Correlation detects linear association

$$r_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}$$

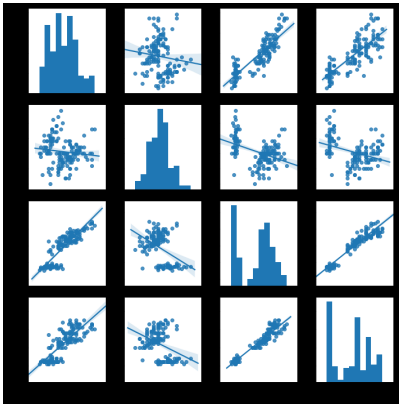
Range

$-1 \leq r \leq 1$ with sign indicating direction.

Zero is limited

Zero correlation does not imply independence.

Correlation reveals redundant measurements



Diagonal panels show individual distributions; other panels show pairs of measurements.

Mutual information captures general dependence

$$I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$$

Meaning

How much observing one variable reduces uncertainty about the other.

Estimation

Finite samples, continuous variables, and tuning choices affect the estimate.

Mutual information can reward an unstable estimate

Rare category

One unusual category appears three times and all three labels are positive.

Small counts

The apparent information may disappear in another sample.

Check

Inspect support counts and selection frequency across folds.

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Chi square tests categorical association

$$\chi^2 = \sum_{r,c} \frac{(O_{rc} - E_{rc})^2}{E_{rc}}$$

Use

Compare observed and expected counts in a contingency table.

Requirements

Inputs should be nonnegative and expected cell counts should not be too small.

Analysis of variance compares group means

$$F = \frac{\text{between group variation}}{\text{within group variation}}$$

Classification

Does a numeric laboratory value have different means across classes

Shape

Outliers, unequal variances, and strong nonnormality can distort the summary.

The exclusive OR problem defeats univariate filters

x1	x2	class
0	0	0
0	1	1
1	0	1
1	1	0

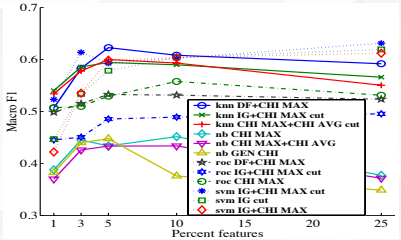
Observation

Neither feature is informative alone, yet the pair determines the class.

Select useful words for text classification

例：特征选择: 文本分类

- 文本分类中的过滤型特征选择（英文中大约有10^5词）



$$\chi^2(x_j) = \sum_y p(y)\chi^2(c, x_j)$$

$$\chi^2(x_j) = \max_y \chi^2(y, x_j)$$

Figure 2: Top 3 feature selection methods for Reuters-21578 (Macro F1)

Monica Rogati, Yiming Yang, High-Performing Feature Selection for Text Classification, CIKM2012

Many tests create false discoveries

$$\mathbb{E}[\text{false positives}] \approx \alpha p$$

Scale

Testing ten thousand noise features at five percent significance can flag hundreds by chance.

Response

Use validation, false discovery control, and stability rather than raw p values alone.

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Selection stability asks whether the same features reappear

- Refit selection on several training resamples or validation folds.
- Record how often each feature is selected.
- Correlated features can substitute for one another.
- Check prediction stability separately from exact feature-set agreement.

Domain review can veto a high score

Question	Example failure	Action
available at prediction	discharge code	remove
stable definition	hospital policy flag	monitor or remove
acceptable use	protected attribute proxy	fairness review
reasonable cost	expensive test	compare benefit with burden



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Forward selection grows a model

1. Start with no selected features.
2. Try adding each remaining candidate.
3. Keep the addition with the best validated improvement.
4. Stop when improvement is too small or the budget is reached.

Greedy search

An early choice can block a better later combination.

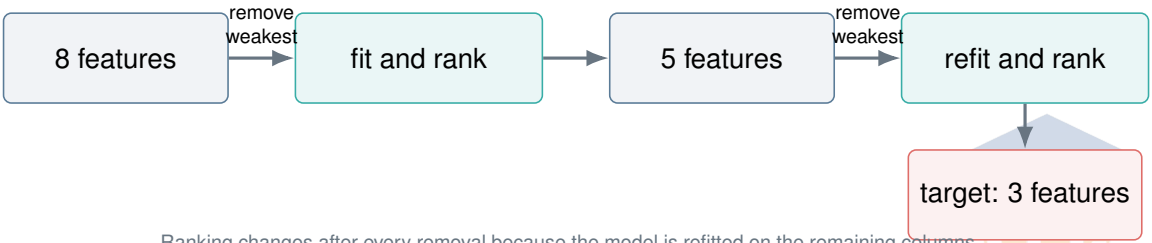
Backward elimination starts with everything

1. Fit the model with all candidate features.
2. Try removing one feature at a time.
3. Drop the feature whose removal hurts validation least.
4. Continue until the stopping rule is met.

Feasibility

Starting with all columns may be impossible when dimension is huge.

Recursive feature elimination follows model weights



Ranking changes after every removal because the model is refitted on the remaining columns.

1. Fit a model on the current feature set.
2. Rank current features by a model specific importance.
3. Remove the weakest feature or group.
4. Refit and repeat until the target size is reached.

Model dependence. A linear weight and a tree importance express different preferences.

Wrapper methods spend models as currency

$$\text{cost} \approx \text{candidate subsets} \times \text{folds} \times \text{model fit time}$$

Scale

Testing 100 additions over 5 folds already requires hundreds of fits.

Budget

Use screening, step size, or early stopping when the search is large.

Nested validation separates search from judgment

Inner loop

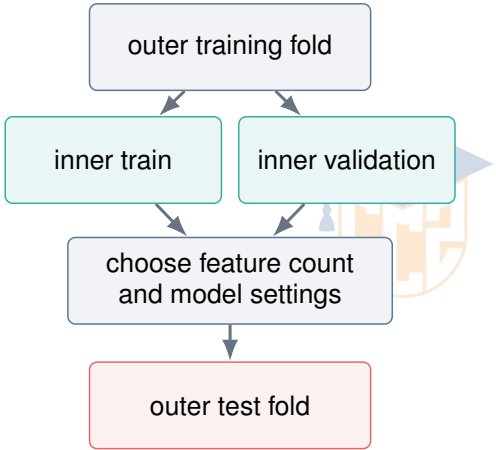
Choose feature count, selection rule, and model settings.

Outer loop

Estimate the performance of that whole search procedure.

Final test

Open the test set only after the procedure has been fixed.



One norm regularization performs embedded selection

$$\min_{w} \widehat{R}(w) + \lambda \sum_{j=1}^{p} |w_j|$$

Effect

Some fitted coefficients become exactly zero.

Scale

Standardize comparable numeric inputs before comparing coefficient shrinkage.

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Lasso can set coefficients to zero

例：嵌入式维度选择

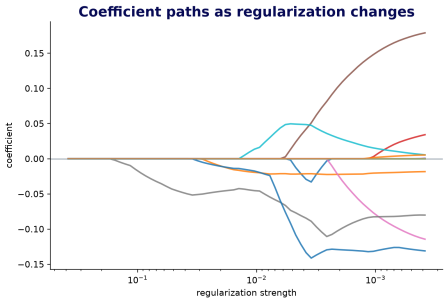
例：商品销量与广告费用（线性回归）

特征	最小二乘线性回归系数	岭回归系数	Lasso系数
TV	3.983944	3.981524	3.921642
radio	2.860230	2.858304	2.806374
newspaper	0.038194	0.038925	0.000000
截距项	13.969091	13.969282	13.972528

Lasso模型选择了2个特征（共3个特征）

$$J(\boldsymbol{w},\lambda)=\sum_{i=1}^{N}L\big(y_i,f(\boldsymbol{x}_i,\boldsymbol{w})\big)+\lambda R(\boldsymbol{w})\ =\|\boldsymbol{y}-\boldsymbol{X}\boldsymbol{w}\|_2^2+\lambda\|\boldsymbol{w}\|_1$$

Lasso paths show entry order and instability



Read the path. Each curve traces one coefficient as regularization changes.

Strong penalty. Most curves remain at zero.

Interpretation. Entry order can change when correlated features or samples change.

Trees select features through splits

Embedded behavior

A feature not used in any split has zero split based importance.

Interaction

A feature can become useful only after an earlier branch conditions the data.

Instability

Correlated predictors may substitute for one another across trees.

Impurity importance can favor many cut points

$$I_j = \sum_{t:\, v(t)=j} \frac{n_t}{n} \Delta i_t$$

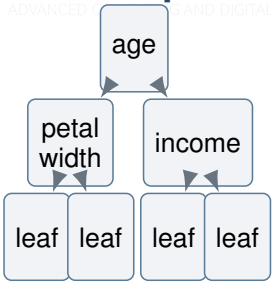
Meaning

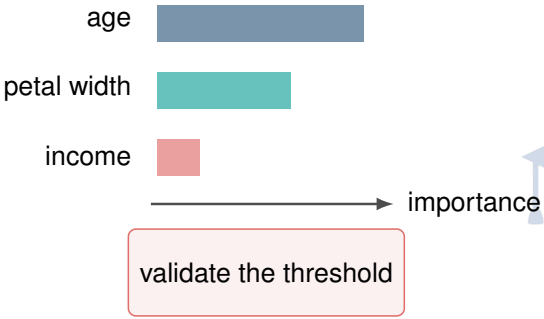
Add weighted impurity decreases from all splits using feature j.

Bias

Continuous or high cardinality features offer more candidate cut points.

Tree selection keeps variables with validated gain





1. Fit the tree model inside a training fold.
2. Estimate feature importance without opening the validation labels.
3. Apply a threshold or select a count.
4. Evaluate the entire pipeline on the held out fold.

Threshold search. The importance threshold is another hyperparameter.

Permutation importance asks about held out performance

$$PI_j = \text{score}(X, y) - \text{score}(X_{\pi(j)}, y)$$

Action

Shuffle one feature column while keeping other columns and fitted model fixed.

Correlation

A correlated partner can preserve the signal and hide importance.

Null importance compares signal with chance

1. Break the label relationship by permutation.
2. Repeat the same importance calculation.
3. Build a null distribution for each feature.
4. Keep candidates whose real importance consistently exceeds chance.

Cost

Repeated permutations can be computationally expensive.

One program can compare selection families

```
from sklearn.datasets import load_breast_cancer
from sklearn.feature_selection import SelectKBest, mutual_info_classif
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import cross_val_score
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler

X, y = load_breast_cancer(return_X_y=True)
pipe = make_pipeline(SelectKBest(mutual_info_classif, k=10),
                     StandardScaler(),
                     LogisticRegression(max_iter=3000))
print(cross_val_score(pipe, X, y, cv=5).mean())
```

Boundary

Selection and scaling are refitted inside every fold.

A clinical panel turns selection into a decision

Candidate	Predictive evidence	Operational concern
routine vital sign	moderate and stable	cheap and always available
special laboratory test	slightly stronger	slow and costly
post treatment code	extremely strong	unavailable at prediction



Decision

Remove the leaked code and compare the first two under a cost aware objective.

Use one fitted pipeline from raw input to prediction

- Raw row → imputation → encoding/scaling → selection → model.
- Fit every learned step on the training portion.
- Keep column order and fitted parameters for new rows.
- Compare the full pipeline using held-out predictions.

One raw row becomes a model vector

Stage	Representation
raw	age 67, city Nanjing, time 23:00, income missing
transform	scaled age, one hot city, cyclic hour, imputed income, missing flag
construct	account age and recent activity ratio
select	retain validated and available columns



Find the mistakes in a preprocessing script

- The mean is calculated before splitting the dataset.
- Features are selected using all labels before cross-validation.
- The test set is repeatedly used to choose preprocessing settings.
- Explain how to fix each step.

A single pipeline repairs the information flow

```
from sklearn.feature_selection import SelectKBest
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import make_pipeline

model = make_pipeline(
    prep,
    SelectKBest(k=3),
    LogisticRegression(max_iter=3000)
)
model.fit(raw_train, raw_y_train)
print(model.score(raw_test, raw_y_test))
```

Meaning

Each learned feature step is fitted from the training rows before the test rows are transformed.

Models react differently to the same representation

Representation issue	Linear model	Tree model
feature scale	changes penalty and optimization	usually little effect on split order
monotonic transform	changes linear shape	often preserves order based splits
one hot encoding	creates class specific coefficients	enables categorical branches through indicators
irrelevant columns	can inflate variance	offer extra noisy split candidates

A deployment gate checks more than accuracy

Gate	Question	Evidence
availability	is every feature present at prediction	timestamp and source audit
consistency	does online computation match training	shared code and tests
stability	do distributions and selections persist	time and subgroup monitoring
value	does the feature improve the decision	validated utility and cost

Explain the method, calculation, and result

Open book prompt

A churn model jumps from 0.72 to 0.96 after adding last contact outcome and selecting 30 of 3,000 columns.

1. Draw the prediction timeline.
2. Audit how selection was fitted.
3. Propose a nested evaluation.
4. State one operational decision.

Weak answer

Listing definitions without diagnosing the score jump does not solve the case.

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Check your understanding

- Distinguish feature construction, extraction, and selection.
- Choose scaling and encoding for a mixed data table.
- Explain how target encoding and selection can leak labels.
- Place preprocessing and feature selection inside cross-validation.